

NALFP — Research Summary

Track #1: Crypto Factor Rebalancing Strategy · Artemis Analytics
Quant Competition · May 2026

Thesis: Crypto cross-sectional returns are better predicted by the *interaction* of network community structure and factor pricing than by either source alone. We build a weekly long/short portfolio that adapts between network and latent-factor signals based on their out-of-sample predictive accuracy.

Strategy Architecture (3 Pillars)

Pillar	Method	Output
1 — Network	Rolling 12-wk Spearman correlation → MST → Louvain communities (7–11/wk, mean entropy 2.18)	<code>within_cluster_mom</code> , <code>cross_cluster_rel</code> , <code>cluster_id</code>
2 — Factor Pricing	9 long/short factor portfolios → Giglio–Xiu (2021) 3-pass with $K_{\text{hidden}}=3$ latent factors; walk-forward $\hat{\lambda}$ refit every 4 wks	$E_{gx} = \beta' \hat{\lambda}$ per asset per week
3 — Adaptive Blend	IC-weighted: $E_{final} = w_{net} \cdot z(E_{net}) + (1 - w_{net}) \cdot z(E_{gx})$; w_{net} = rolling 8-wk OOS Spearman IC ratio	Blended expected return ranking

Portfolio: Long top quintile / short bottom quintile · inverse-vol weights · dollar-neutral · 5% max per name · 40% cluster cap · 30% turnover budget · 15% vol target (3× lev cap). Median 16 long + 16 short names/week.

Factor Zoo — Key Results (Full Sample & GX Cross-Section)

Factor	Ann. Ret	Sharpe	$\hat{\lambda}_{full}$ (%/wk)	t(full)	Verdict
SMBC (size)	+93.7%	+3.54	+0.59	+1.62	Priced (border- line)
NetRel (cross- cluster rota- tion)	+74.8%	+2.24	+1.12	+2.38	Priced
NetMom (within- cluster mom)	+30.4%	+1.33	+0.76	+2.42	Priced
MomC (momen- tum)	+40.0%	+1.09	+0.96	+1.74	Priced
VolC (low-vol)	+10.0%	+0.22	+0.99	+3.24	Priced (highest t)
RC (market)	−32.4%	−0.78	−0.25	−1.13	Not priced (regime- dependent)
FunC (fund. yield)	+26.6%	+0.85	−0.12	−0.32	Not priced (sparse coverage)

GX hidden factors H1–H3: individually insignificant ($|t| < 1$). Five of seven observed factors survive $|t| \geq 1.65$ in the observed-only model; four survive after latent adjustment.

OOS Backtest Results (24 Weeks, Walk-Forward)

	NALFP	NALFP (no cap)	EW Mom Long	plus_lat
Ann. Return	−14.8%	−17.4%	+58.1%	+129.0%
Ann. Vol	10.3%	10.3%	43.0%	34.4%
Sharpe (OOS)	−1.44	−1.69	+1.35	+3.75

	NALFP	NALFP (no cap)	EW Mom Long	plus_lat
Sharpe	[−3.45,	[−3.66,	[−2.08,	[0.51, +8.89]
95% CI	+1.18]	+1.11]	+6.61]	
Max DD	−6.6%	−7.8%	−24.8%	−7.4%
Turnover/wk	7.4%	7.4%	33.4%	65.1%

Training Sharpe: +2.30. OOS blend weight on network: $\bar{w}_{net} = 0.25$ (network IC negative OOS $\rightarrow$ blend tilts heavily to GX). Cluster cap improves OOS Sharpe by +0.25 and reduces drawdown by 1.2 pp.

Honest Assessment

What worked: (1) Factor identification is statistically grounded — four factors survive latent-adjustment pricing. (2) The network pillar is operationally useful — the cluster cap reduces drawdown even when its cross-sectional signal is negative OOS. (3) Vol targeting delivers single-digit volatility and −6.6% max drawdown vs. −24.8% for the momentum benchmark.

What failed: (1) OOS Sharpe is −1.44 with a CI that crosses zero — the strategy does not earn positive risk-adjusted returns in this sample. (2) Network signal IC is negative OOS (−0.014); the blend defaults to GX. (3) Factor premia reverse between train and test: the strong small-cap premium in training (SMBC Sharpe +3.54) does not persist.

What would break this: A regime shift flipping SMBC or MomC signs out-of-sample — exactly what occurred in the OOS window. Whether $\hat{\lambda}$ generalises is an empirical question unanswerable with 24 OOS weeks.

Structural limits: 52-wk sample / 24-wk OOS · K_hidden = 3 capped, not data-driven · TVLC & SupC excluded for sparse coverage · 10 bps cost model optimistic for tail names · no funding-rate or perp-basis data.